Resilience Academy

SDI Training for ICT Professionals

Kobo to CRD Integration

From Field Data to the National SDI – No Programming Required

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Day 3 | 30 Minutes Lecture | crd.resilienceacademy.ac.tz

Resilience Academy

The Data Flow: Field to National SDI

KoboToolbox $\xrightarrow{\text{export}}$ **CSV** $\xrightarrow{\text{open}}$ **QGIS** $\xrightarrow{\text{save as}}$ **GeoJSON** $\xrightarrow{\text{upload}}$ **CRD**

Step	Action	Tool
1	Export data as CSV	KoboToolbox (browser)
2	Save CSV to computer	Your Downloads folder
3	Open CSV, verify points on map	QGIS
4	Export as GeoJSON	QGIS
5	Upload GeoJSON to CRD	CRD (browser)
6	View on dashboard with imagery	CRD (browser)

What This Pipeline Does

Takes flood observation points collected on phones in the field and puts them on the Climate Risk Database where anyone with access can view them overlaid on satellite imagery.

No Programming Required

Every step uses a **graphical interface** – clicking buttons and menus. No Python, no command line, no code. If you can use a web browser and QGIS, you can do this.

Step 1: Export from KoboToolbox as CSV

In KoboToolbox (your web browser):

1. Log in to KoboToolbox
2. Open the **Tanga Flood Survey** form
3. Click the **"Data"** tab at the top
4. Click **"Downloads"** in the left sidebar
5. Under "Export type", select **"CSV (legacy)"**
6. Click the blue **"Export"** button
7. Wait a few seconds for the file to be generated
8. Click **"Download"** to save the file to your computer

What you get:

A file called something like

What is Inside the CSV?

A spreadsheet with one row per submission:

Column	Example
observer	Amina J.
date	2026-03-09
_geolocation_latitude	-5.0732
_geolocation_longitude	39.0993
flood_depth_cm	45
severity	High
damage	Roads; Buildings

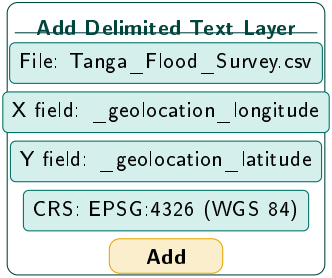
Key Columns

The **latitude** and **longitude** columns

Step 2: Open the CSV in QGIS

In QGIS (on your computer):

1. Open **QGIS** on your computer
2. Go to menu: **Layer** → **Add Layer** → **Add Delimited Text Layer...**
3. In the dialog that opens:
   - **File name:** Click "Browse" and find your downloaded CSV file
   - **File format:** Select "CSV"
   - **Geometry definition:** Select "Point coordinates"
   - **X field:** Choose `_geolocation_longitude`
   - **Y field:** Choose `_geolocation_latitude`
   - **CRS:** Leave as EPSG:4326 (WGS 84)
4. Click **"Add"**



Important

Make sure X = **longitude** and Y = **latitude**. Mixing them up puts your points in the wrong place!

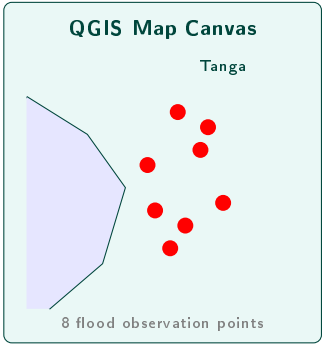
Step 3: Verify Your Points on the Map

Check That Your Data Looks Right

After adding the CSV, you should see dots on the QGIS map canvas. Each dot represents one flood observation from the field.

What to verify:

- ✓ Points appear in the **Tanga area** (not in the ocean, not in another country)
- ✓ The **number of points** matches the number of submissions in KoboToolbox
- ✓ Click on a point with the **Identify tool** (the "i" button) to see its attributes
- ✓ Check that observer name, date, severity, and



If Points Look Wrong

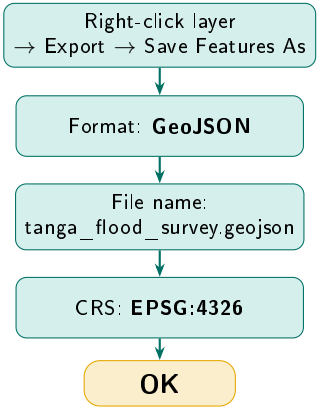
Check that you selected the correct latitude and longitude columns in Step 2. Re-add the layer if needed.

Step 4: Export from QGIS as GeoJSON

Why GeoJSON?
The CRD (Climate Risk Database) accepts **GeoJSON** files for upload. GeoJSON is a standard format that stores both the location (coordinates) and the attributes (observer, severity, etc.) together.

How to export:

1. In QGIS, **right-click** on the flood survey layer in the Layers panel
2. Select **"Export"** → **"Save Features As..."**
3. In the dialog:
   - **Format:** Choose "GeoJSON"
   - **File name:** Click "Browse" and choose where to save it (e.g., Desktop)
   - Name it: `tanga_flood_survey.geojson`



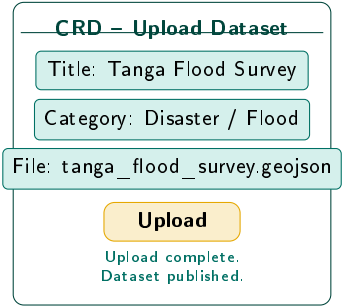
File Size

A GeoJSON file with a few hundred

Step 5: Upload GeoJSON to the CRD

In the CRD web interface (your browser):

1. Open your browser and go to **crd.resilienceacademy.ac.tz**
2. Log in with your CRD account
3. Navigate to the **Upload** or **Data Management** section
4. Click **"Upload Dataset"** (or similar button)
5. In the upload dialog:
   - **Title:** "Tanga Flood Survey – March 2026"
   - **Category:** Disaster / Flood
   - **File:** Click "Browse" and select your `tanga_flood_survey.geojson`
6. Click **"Upload"**



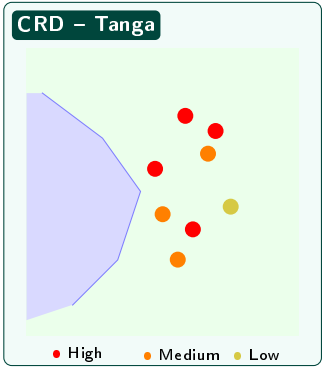
Metadata

Add a description and source information so other users know where this data came from and when it was collected.

Step 6: View Flood Survey Points on the CRD Map

Your Data is Now Live on the Dashboard

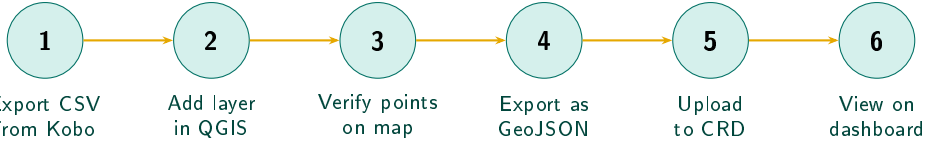
After uploading, navigate to the **Map** view in the CRD. You will see your flood survey points overlaid on Tanga satellite imagery.



What you can do in the CRD:

- **Zoom** to the Tanga area to see all flood points
- **Click** on a point to see its details (observer, severity, depth, date)
- **Toggle** satellite imagery on/off to compare with flood locations
- **Overlay** with other datasets (e.g., building

Why This Matters: Field Data → National SDI in 6 Clicks



What You Achieved Today

- Collected GPS-tagged flood data on phones
- Exported it from KoboToolbox
- Verified it in QGIS
- Published it to the national CRD
- **All without writing a single line of**

Why This Changes Everything

- **Speed:** Field data reaches decision-makers the same day
- **Accuracy:** GPS removes guesswork about locations
- **Accessibility:** Anyone with CRD access can see the data